

Introduction

Gun violence in the United States is a persistent public health crisis that disproportionately affects young people of color in both urban and rural areas. Firearm homicides are especially concentrated in large metropolitan regions. In 2018 alone, 88.5% of all firearm homicides (12,357 deaths) occurred in metropolitan counties (CDC, 2020). That same year, non-Hispanic Black individuals aged 15 to 34 accounted for over 62% of intentional firearm homicide victims, despite representing only 14.8% of the population (Kegler, Dahlberg, & Mercy, 2018).

While some scholars argue that policing and punishment can reduce gun violence (Braga & Cook, 2023), others caution that aggressive enforcement strategies — such as stop-and-frisk or high incarceration rates — often carry serious social costs for communities that are already disproportionately affected by gun violence (Harcourt, 2005; Raphael & Stoll, 2013). One significant barrier to reducing firearm violence is the chronic underreporting of gunfire incidents. Studies suggest that only about 20% of shootings are reported to law enforcement (Irvin-Erickson et al., 2016). In response, cities across the country have turned to gunshot detection technology (GDT) — most notably, ShotSpotter — to improve real-time awareness of gunfire and overcome reliance on inconsistent civilian reporting (Choi, Librett, & Collins, 2014).

ShotSpotter in Chicago

ShotSpotter, developed in 1996, is now used in over 100 U.S. cities. The technology uses acoustic sensors and triangulation to detect gunshots and relay their locations to police departments. After an alert is triggered, a trained ShotSpotter employee (an “acoustic expert”) confirms whether the sound is indeed a gunshot, and if so, the alert is forwarded to law enforcement (Gatens & Reichert, 2019). Originally adapted from earthquake detection systems,

the goal of ShotSpotter is to provide fast, consistent, and data-driven responses to gun violence, especially in areas where 911 call data is unreliable.

ShotSpotter was first piloted in Chicago in 2012 in two small zones within the 7th (Englewood) and 11th (Harrison) police districts. In 2016, the system expanded to cover the entirety of both districts. The 2017 expansion across multiple districts was part of a broader policing strategy under then-Mayor Rahm Emanuel, who positioned ShotSpotter as a central tool in the city's gun violence reduction plan. The contract with SoundThinking reflected a significant public investment in surveillance technology. Following this expansion, the City of Chicago signed a three-year, \$3 million contract with SoundThinking (formerly ShotSpotter) in 2018 to widen coverage further (Office of Inspector General, 2021).

ShotSpotter has also faced public criticism, including questions about its accuracy, transparency, and disproportionate targeting of Black and Brown communities. Investigative reports and community organizations have raised concerns that the technology may reinforce over-policing in historically marginalized areas, while failing to address root causes of violence (Mac, 2021; Inspector General, 2021). In 2023, Mayor Brandon Johnson announced plans to end the city's ShotSpotter contract, citing concerns about the system's reliability and community impact. While some city council members supported this decision, others, particularly from majority-Black wards, argued that ShotSpotter still holds value if it can help save lives. These concerns make it essential to evaluate not only whether ShotSpotter detects gunfire, but also whether it supports equitable public safety outcomes.

Research on ShotSpotter's effectiveness has yielded mixed results. Some studies suggest it may reduce response times or increase arrests, while others find little to no impact on actual crime rates or community outcomes. While most evaluations focus on operational outcomes like

arrests or pre-hospital time, few have critically examined how ShotSpotter interacts with broader patterns of community disinvestment and structural inequality.

Framing Our Contribution

This paper fills that gap by asking:

Do neighborhoods with sustained high ShotSpotter alerts share characteristics of long-term economic disinvestment?

We focus on six high-violence neighborhoods in Chicago between 2017 and 2024. Using spatial and temporal data from ShotSpotter alerts and police-reported homicides, paired with neighborhood-level indicators such as poverty, housing vacancy, and homeownership, we explore how persistent gunfire exposure relates to broader conditions of structural inequality. In doing so, we aim to evaluate not just whether ShotSpotter works as a detection tool, but also what its use reveals about where violence concentrates, why it persists, and what kinds of responses are truly needed.

As cities debate whether to renew or expand ShotSpotter contracts, understanding its role within the broader ecosystem of violence, disinvestment, and public safety is more important than ever. This research contributes to ongoing policy debates about whether public safety should be driven by technological surveillance or sustained social and economic investment in the communities most impacted by gun violence. For this reason, we evaluate ShotSpotter not as a stand-alone solution, but as one part of a much larger conversation about equity, investment, and justice in communities most impacted by gun violence.

Data Description

This study draws on multiple data sources to examine the connection between ShotSpotter alerts, fatal violence, and socioeconomic conditions in Chicago's most affected

neighborhoods. The analysis focuses on the neighborhoods of Austin, Englewood, North Lawndale, South Shore, West Englewood, and Humboldt Park. The primary source of gunfire detection data comes from the City of Chicago’s ShotSpotter alert system (2017–2024), which records real-time incidents of gunfire across designated violence reduction zones. Homicide data from the "*Crimes - 2001 to Present*" dataset was used to examine fatal violence trends alongside ShotSpotter alerts. This data was aggregated by neighborhood and month.

To evaluate long-term structural disadvantage, the study incorporates neighborhood-level socioeconomic indicators from the City of Chicago’s 2008–2012 census dataset. These include the percentage of households below poverty, unemployment rates, the percentage of adults without a high school diploma, and a composite hardship index. **Figure 1** presents the selected socioeconomic indicators for the six neighborhoods. The data show high poverty and hardship scores in areas like Englewood, North Lawndale, and West Englewood, suggesting a strong foundation of disinvestment preceding the period of ShotSpotter deployment.

Census Data - Selected socioeconomic indicators in Chicago, 2008 – 2012							
Community Area Name	Percent Of Housing Crowded	Percent Households Below Poverty	Percent Aged 16+ Unemployed	Percent Aged 25+ Without High School Diploma	Percent Aged Under 18 Or Over 64	Per Capita Income	Hardship Index
Austin	6.3	28.6	22.6	24.4	37.9	15957	73.0
West Garfield Park	9.4	41.7	25.8	24.5	43.6	10934	92.0
North Lawndale	7.4	43.1	21.2	27.6	42.7	12034	87.0
South Shore	2.8	31.1	20.0	14.0	35.7	19398	55.0
Englewood	3.8	46.6	28.0	28.5	42.5	11888	94.0
Auburn Gresham	4.0	27.6	28.3	18.5	41.9	15528	74.0

Figure 1: Socioeconomic Indicators by Neighborhood (2008–2012)

Table comparing key socioeconomic conditions in six Chicago neighborhoods before ShotSpotter implementation. Indicators include poverty rate, unemployment, education, income, and hardship index.

Spatial patterns were also analyzed using GIS-based maps of homicide concentrations before and after ShotSpotter implementation. **Figures 2, 3, and 4** display neighborhood-level shifts in violence over time, highlighting the areas that remained persistently high in shootings.

Chicago Most Violent Neighborhoods 2009-01-03 - 2017-01-13

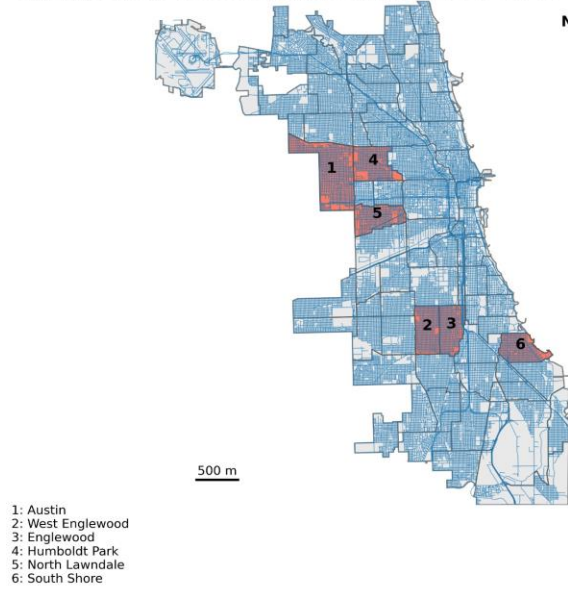


Figure 2: Most Violent Neighborhoods in Chicago (2009- 2017)

Map identifying the most violent neighborhoods based on homicides between 2009 and early 2017, before ShotSpotter was deployed.

Chicago Most Violent Neighborhoods 2017-01-13 - 2024-06-29

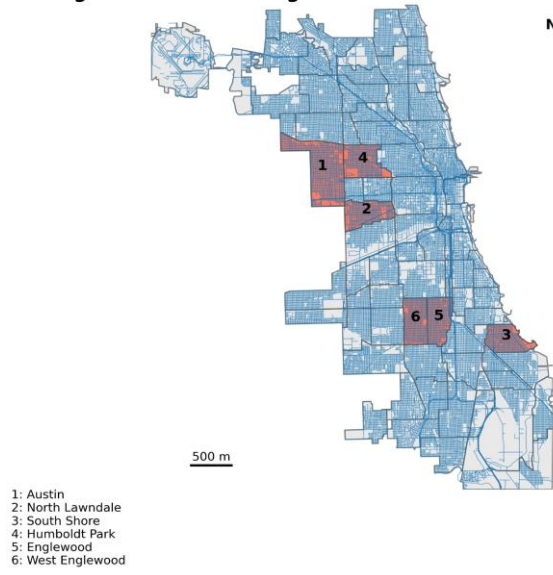


Figure 3: Most Violent Neighborhoods in Chicago (2017-2024)

Map showing homicide hotspots between 2017 and 2024 during the period of ShotSpotter implementation

Chicago Most Violent Neighborhoods 2009-01-01 - 2024-06-29

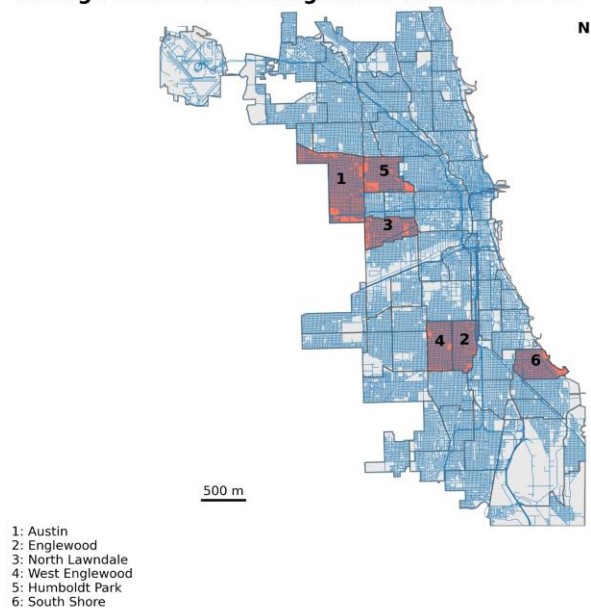


Figure 4: Cumulative Homicides Across Chicago (2009-2024)

Map highlighting persistent violent neighborhoods by aggregating homicides from 2009 to 2024.

Figure 5 compares total homicides before and during ShotSpotter implementation in each neighborhood. This allows for a direct neighborhood-level assessment of whether homicides rose, declined, or stayed flat following the introduction of the technology.

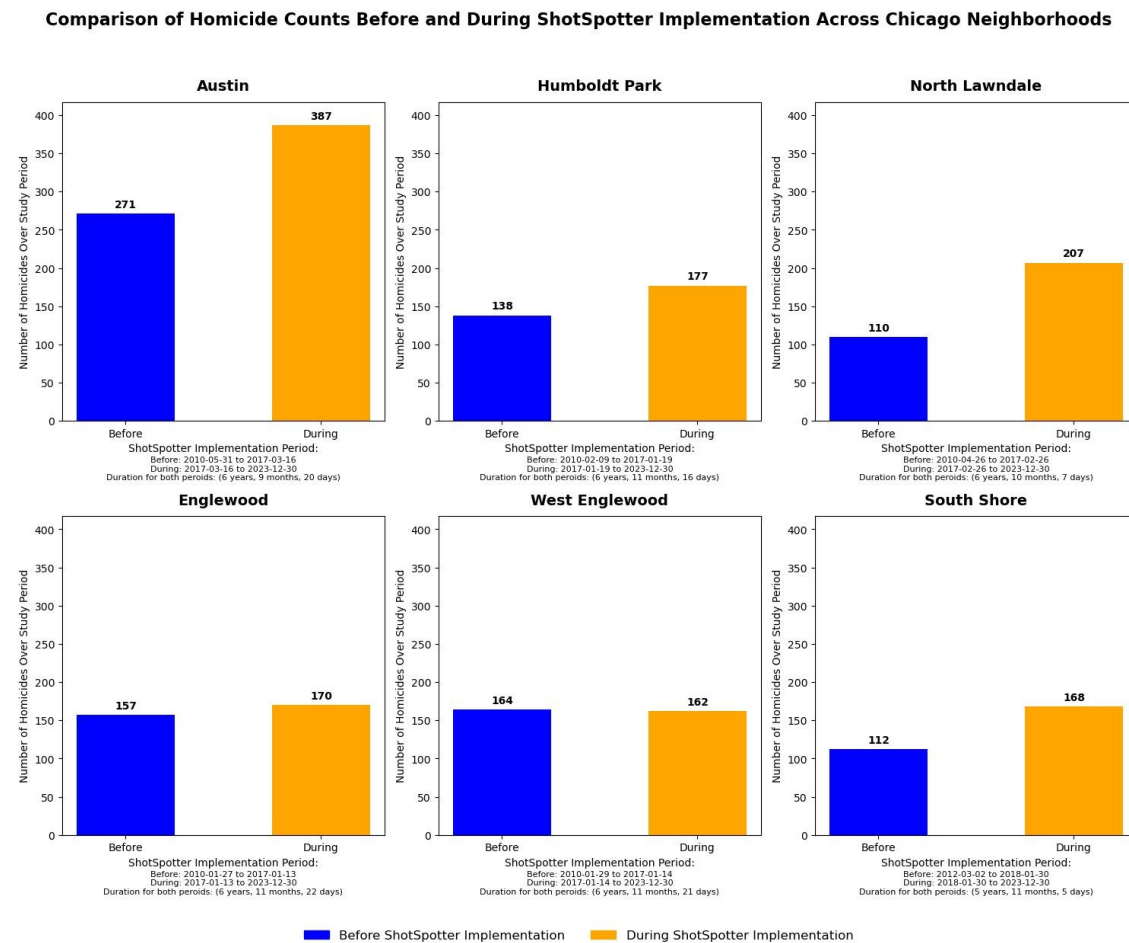


Figure 5: Homicides before vs. During ShotSpotter Implementation

Bar chart comparing total homicides in each neighborhood before and during ShotSpotter implementation. Useful for evaluating the overall change in fatal violence.

Figure 6 adds more granularity to this analysis by showing grid-level maps of homicides (pre- and post-2017) and ShotSpotter alert intensity for each neighborhood. This visual illustrates

the overlap or divergence in spatial concentration between gunfire and fatal shootings, helping us understand whether ShotSpotter zones align with actual violence patterns.

Empirical Strategy

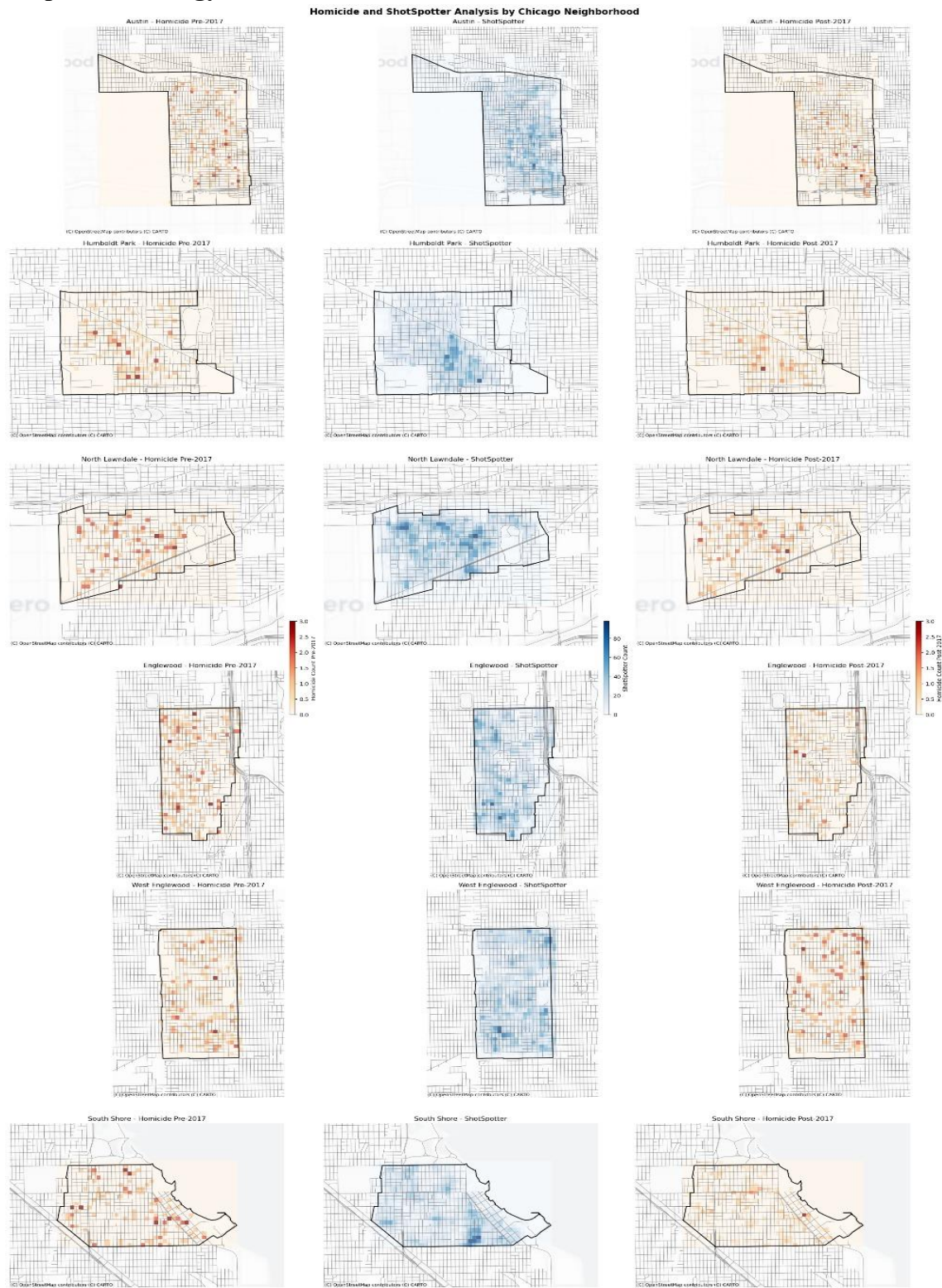


Figure 6: Neighborhood Heatmaps: ShotSpotter vs. Homicides (Pre/Post)

Grid of neighborhood maps comparing spatial distribution of homicides (pre- and post-2017) with ShotSpotter alert density. Helps identify spatial overlap between detected gunfire and actual fatal violence. (See code for blown up image)

The analysis is descriptive and comparative. It involves: (1) tracking changes in monthly ShotSpotter alert counts from 2017 to 2024, (2) comparing those alerts to monthly homicide counts in the same neighborhoods, and (3) layering in 2008–2012 socioeconomic indicators to assess structural context.

Time-series graphs and correlation scatterplots were used to visualize the association between alerts and homicides. **Figure 7** plots alerts and homicides over time in each neighborhood, separated by pre- and post-installation periods. Trend lines were added to highlight trajectory shifts.

Monthly Homicides and ShotSpotter Alerts Across Chicago Neighborhoods with Trend Lines for Before and During Periods

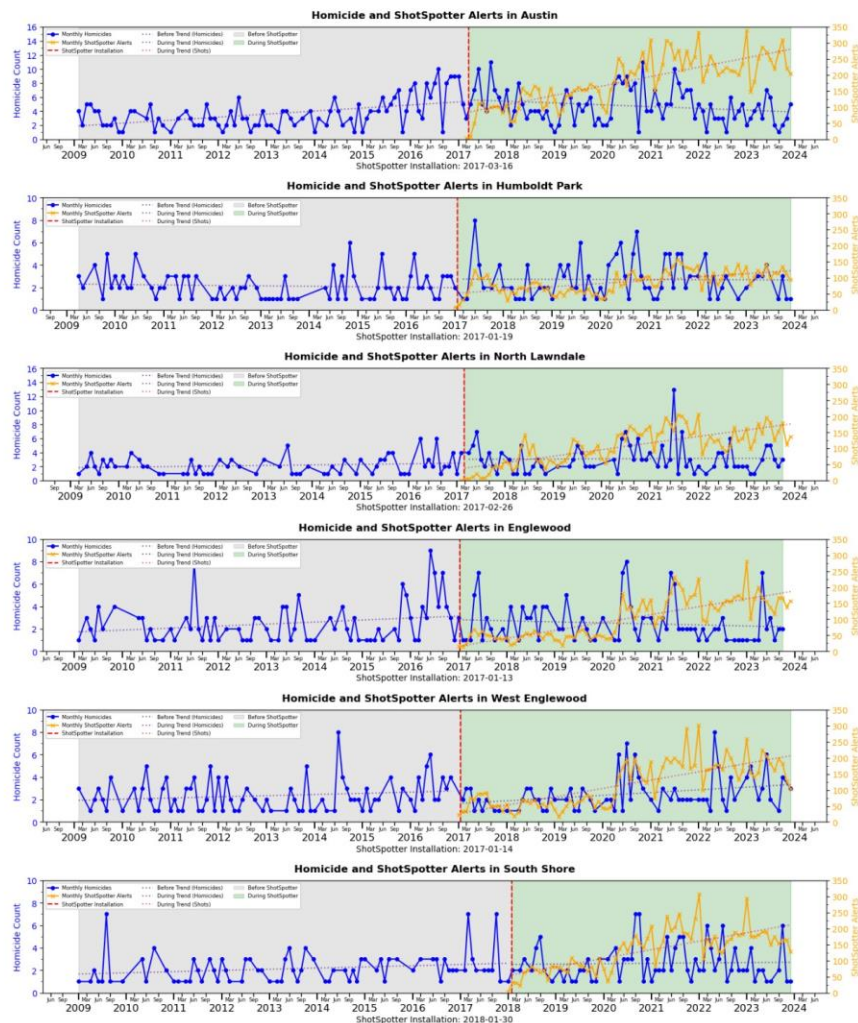


Figure 7: ShotSpotter Alerts and Homicides Over Time by Neighborhood (2009–2024)

Line graphs showing monthly trends in ShotSpotter alerts and homicides in each neighborhood from 2009–2024, separated by before and after implementation, with trend lines

A second visualization, **Figure 8**, compares mean ShotSpotter alerts and mean homicides by month across all six neighborhoods. This captures seasonality and lag patterns in gunfire and fatal incidents.

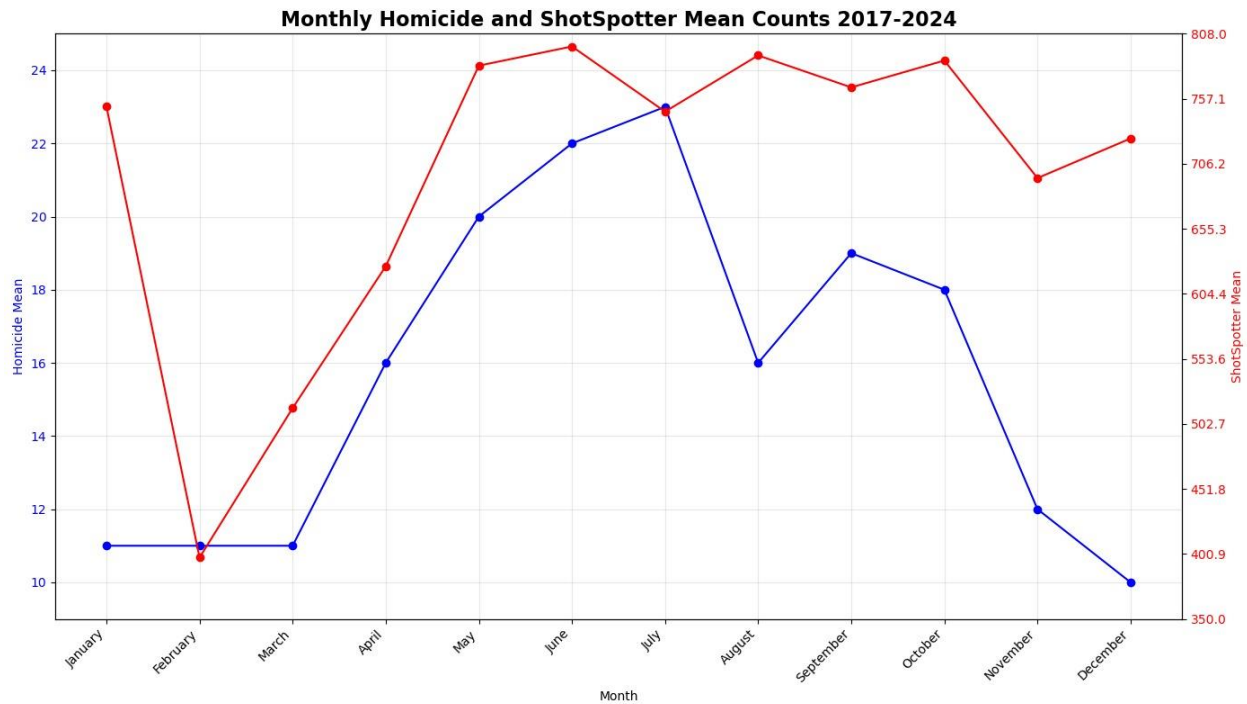


Figure 8: Monthly Mean ShotSpotter Alerts vs. Homicides (2017–2024)

Graph showing average monthly ShotSpotter alerts and homicides from 2017–2024 across all neighborhoods. Highlights seasonality, with summer months peaking in both alerts and fatal incidents

Figure 9 shows the correlation between monthly ShotSpotter alerts and homicides per neighborhood. Each panel includes a trend line and statistical correlation results. This helps test whether increases in alerts are associated with increases in lethal violence.

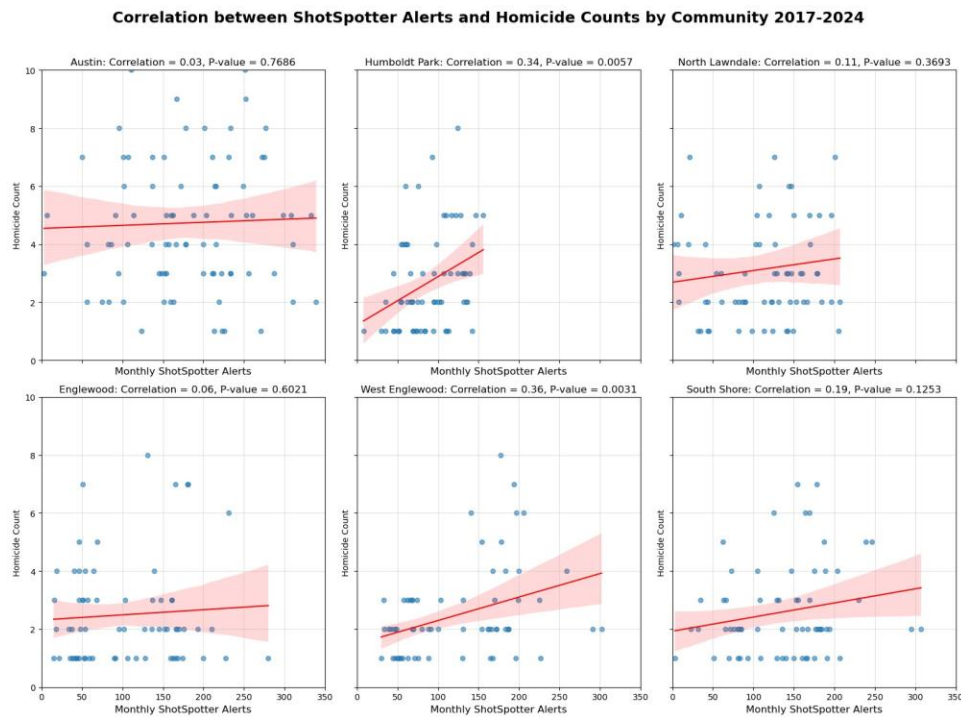


Figure 9: Correlation Between Alerts and Homicides (2017–2024)

Scatterplots for each neighborhood showing the correlation between monthly ShotSpotter alerts and homicides. Regression lines and p-values included to assess statistical significance.

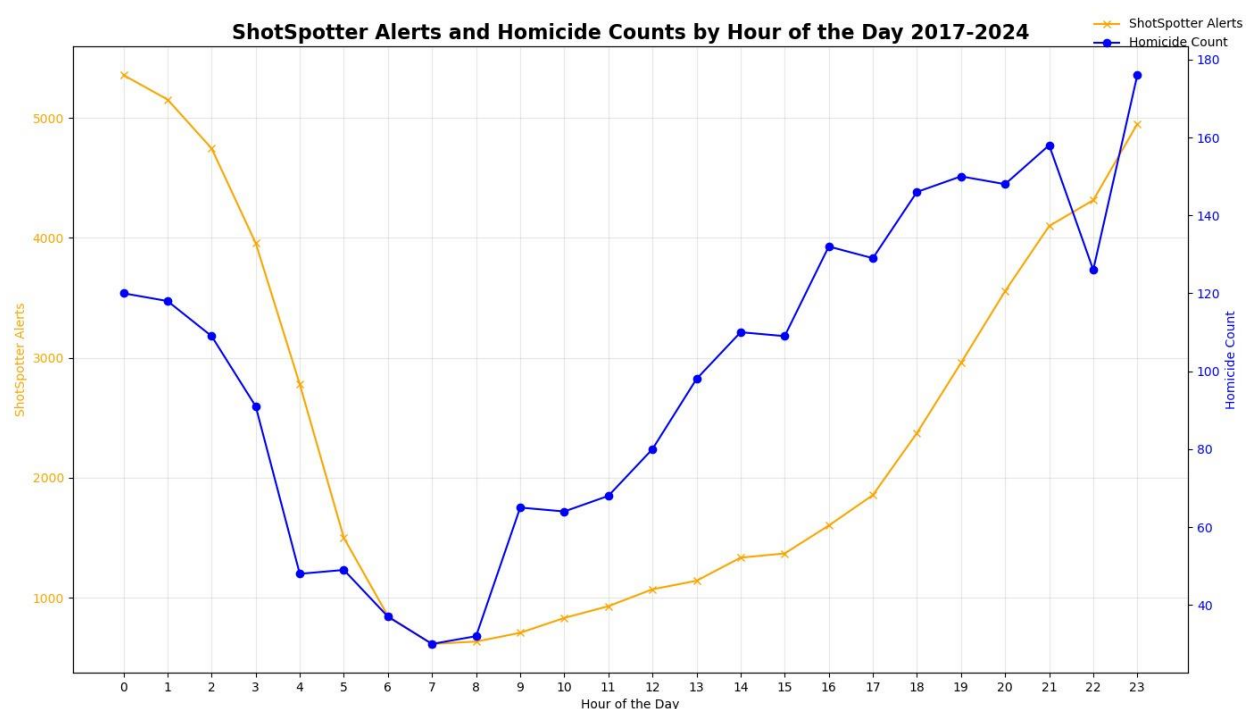
Results

The results reveal that ShotSpotter alerts increased substantially after implementation across all six neighborhoods. However, the change in homicide rates was inconsistent. For example, West Englewood and Humboldt Park showed statistically significant correlations between alerts and homicides, while Austin, Englewood, and North Lawndale showed near-zero or weak correlations. **Figure 9** shows these differences clearly: West Englewood ($r = 0.36$, $p = 0.0031$) and Humboldt Park ($r = 0.34$, $p = 0.0057$) both demonstrate moderately strong relationships. South Shore also had a positive correlation ($r = 0.19$), though not statistically significant. The remaining neighborhoods exhibit little to no meaningful connection between

ShotSpotter data and homicides. **Figure 8** shows that both alerts and homicides tend to spike during the summer months, particularly from May through August. This seasonal pattern aligns with broader urban violence trends. **Figure 10** examines ShotSpotter alerts and homicide counts by hour of the day. Most gunfire and lethal incidents occur late at night, peaking between 10:00 PM and midnight.

Figure 10. ShotSpotter Alerts and Homicides by Hour of the Day (2017–2024)

Line graph comparing hourly counts of ShotSpotter alerts and homicides across all neighborhoods. Both metrics rise in the evening and peak around midnight.



The spatial maps in **Figures 2–4** confirm that the neighborhoods studied remained among the most violent before and after ShotSpotter was implemented. While some shifts occurred, most of these communities remained persistently affected. **Figure 6** reinforces this by showing overlapping clusters of ShotSpotter alerts and homicides at the sub-neighborhood level. In some

areas, these align closely, while in others, ShotSpotter detects gunfire in areas with few recorded homicides, raising questions about the nature of non-fatal or unreported gun activity.

Discussion

Overall, the findings suggest that more ShotSpotter alerts don't automatically mean more homicides. Most neighborhoods show little to no connection between gunfire alerts and fatal violence. ShotSpotter seems to be better at identifying the presence of gunfire, but it doesn't necessarily capture whether that gunfire results in injury or death. For neighborhoods like West Englewood and Humboldt Park, the stronger correlations suggest something different may be happening, possibly related to gang activity, police presence, or other local dynamics. But for the most part, these alerts alone don't explain why some communities continue to face deadly violence while others don't.

The table of socioeconomic indicators (**Figure 1**) sheds light on why some neighborhoods continue to face higher levels of violence despite similar alert levels. Places like Englewood, North Lawndale, and West Garfield Park have extremely high hardship scores, low income, and poor educational attainment. These are long-standing issues that ShotSpotter alone can't fix. What the graphs and data do show is that surveillance technology is not a substitute for real investment. Gunfire detection can tell us where violence happens, but it can't address the root causes, like housing insecurity, unemployment, and the lack of opportunity. That's why policies aimed at expanding affordable housing, supporting youth employment, and funding community-based violence prevention programs are critical. Programs like READI Chicago or Cure Violence have shown promise in addressing violence through outreach, trauma support, and job readiness — something ShotSpotter alone can't provide.

Conclusion

While this analysis reveals important trends, it has limitations, including possible bias in ShotSpotter coverage and the absence of detailed data on non-lethal incidents or community trust. Still, the patterns point to deeper structural forces behind persistent violence. This paper examined the relationship between ShotSpotter alerts and homicides in six of Chicago's most violent neighborhoods. While alerts increased after installation, they did not consistently lead to changes in homicide trends. Some neighborhoods—West Englewood and Humboldt Park—showed moderate correlations, but others did not. These findings highlight the limits of technology in explaining or reducing violence. Structural indicators like poverty and disinvestment play a far greater role and must be part of any long-term solution. In short, technology might help detect the problem, but it's not going to solve it. Real change requires investment in education, housing, economic opportunity, and community trust.

References

Braga, A. A., & Cook, P. J. (2023). *Policing gun violence: Strategic reforms for controlling our most pressing crime problem*. Oxford University Press

Carr, Jillian, and Jennifer L. Doleac. "The geography, incidence, and underreporting of gun violence: new evidence using ShotSpotter data." *Incidence, and Underreporting of Gun Violence: New Evidence Using Shotspotter Data (April 26, 2016)* (2016).

Centers for Disease Control and Prevention (CDC). (2020). Firearm mortality by state and demographic group—United States, 2018.

<https://www.cdc.gov/nchs/pressroom/states/illinois/il.htm>

Choi, K. S., Librett, M., & Collins, T. J. (2014). An empirical evaluation: gunshot detection system and its effectiveness on police practices. *Police Practice and Research*, 15(1), 48-61.
<https://www.tandfonline.com/doi/full/10.1080/15614263.2013.800671>

Gatens, A., & Reichert, J. (2019). *Police technology: Acoustic gunshot detection systems*. Illinois Criminal Justice Information Authority.

Harcourt, B. E. (2019). *Against prediction: Profiling, policing, and punishing in an actuarial age*. University of Chicago Press.

Irvin-Erickson, Y., Bai, B., Gurvis, A., & Mohr, E. (2016). The effect of gun violence on local economies.

Kegler, S. R. (2018). Firearm homicides and suicides in major metropolitan areas—United States, 2012–2013 and 2015–2016. *MMWR. Morbidity and Mortality Weekly Report*, 67.

Office of Inspector General, City of Chicago. (2021, August 24). *The Chicago Police Department's use of ShotSpotter technology*
<https://www.chicago.gov/content/dam/city/depts/oig/publications/reports-public-safety/2021/CPD-ShotSpotter-Report-8-24-21.pdf>

Raphael, S., & Stoll, M. A. (2013). *Why are so many Americans in prison?*. Russell Sage Foundation.

Thompson, P. S., Lawrence, D. S., & Vigne, N. G. (2020). *Securing officer buy-in when a beneficial technology increases workload: Best practices for implementing gunshot detection technology*. *ACJS Today*, 46(1), 1-7.